SDK User Manual

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1. 编译环境

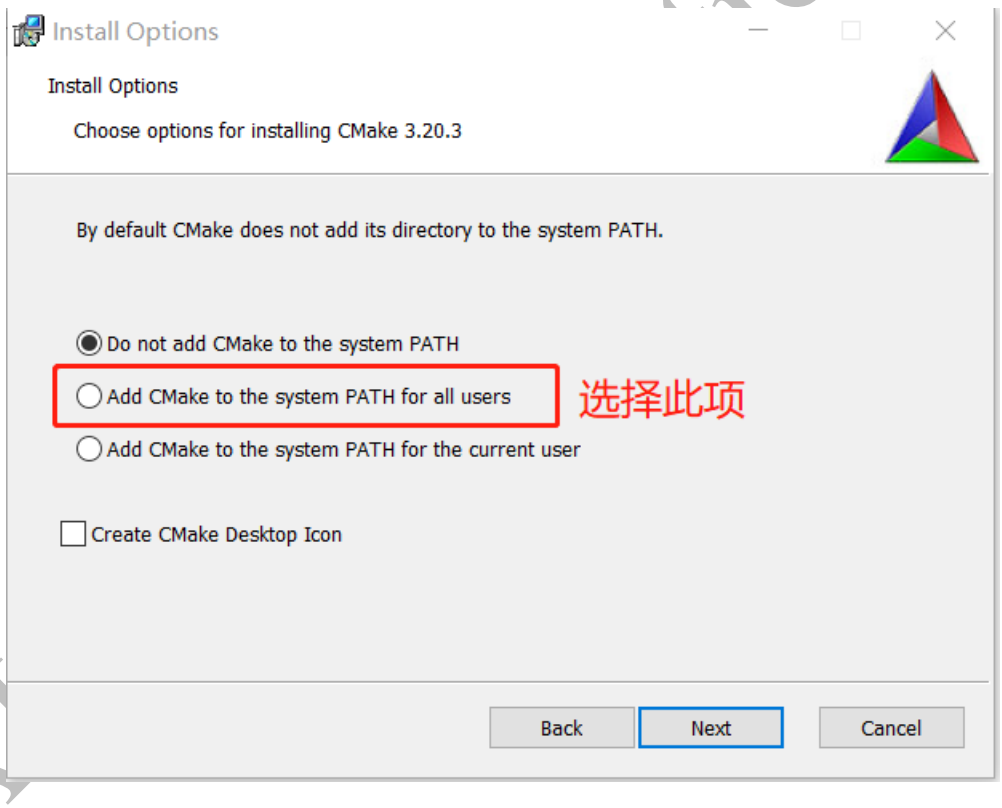
GX3xx SDK 支持 Windows 10 和 Windows7, 推荐使用 windows10。编译需要包含如下工具: CMake、Git for Windows、Python and Ninja.

1.1.CMake

SDK 基于 CMake 编译。CMake 软件版本为 3.21.1 或以上。可从如下 SDK 目录获取 CMake 安装包:

"SDK/tools/cmake/"*

安装时，请勾选如下选项将 CMake 添加到系统环境变量中：



安装完成后，检查 CMake 版本信息：

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```
comuser@ip-AC10022A MINGW64 /p/00_development/comuser/code
$ cmake --version
cmake version 3.21.1

CMake suite maintained and supported by Kitware (kitware.com/cmake).
```

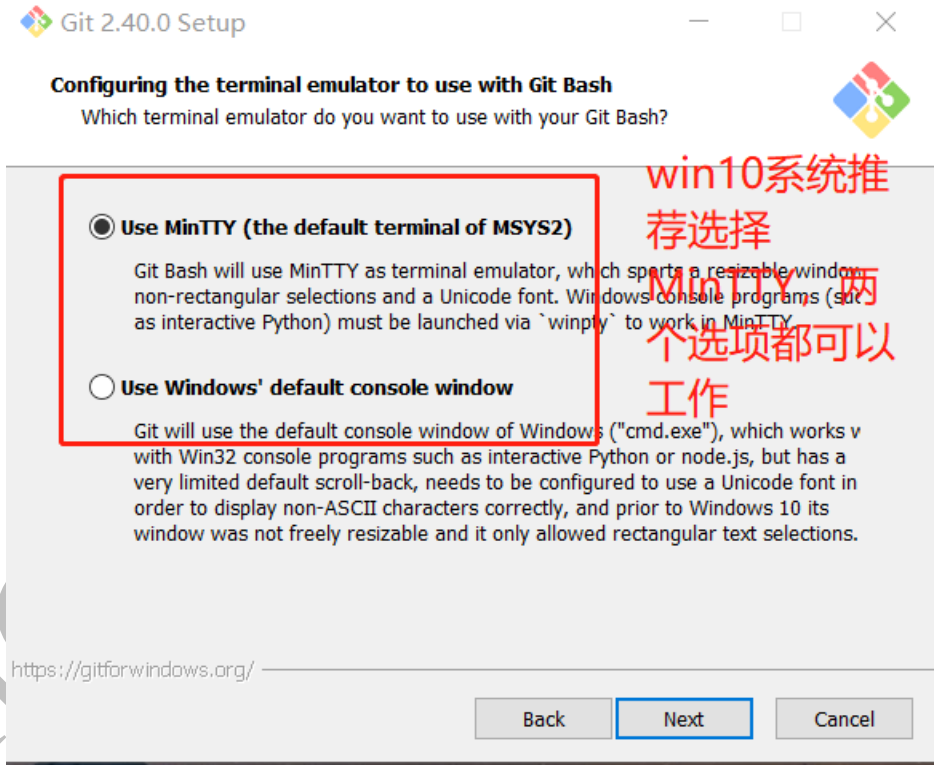
1.2.Git for Windows

Git 工具用做 SDK 代码管理及编译终端，可在如下 SDK 路径中获取其安装包：

“SDK/tools/git/”*

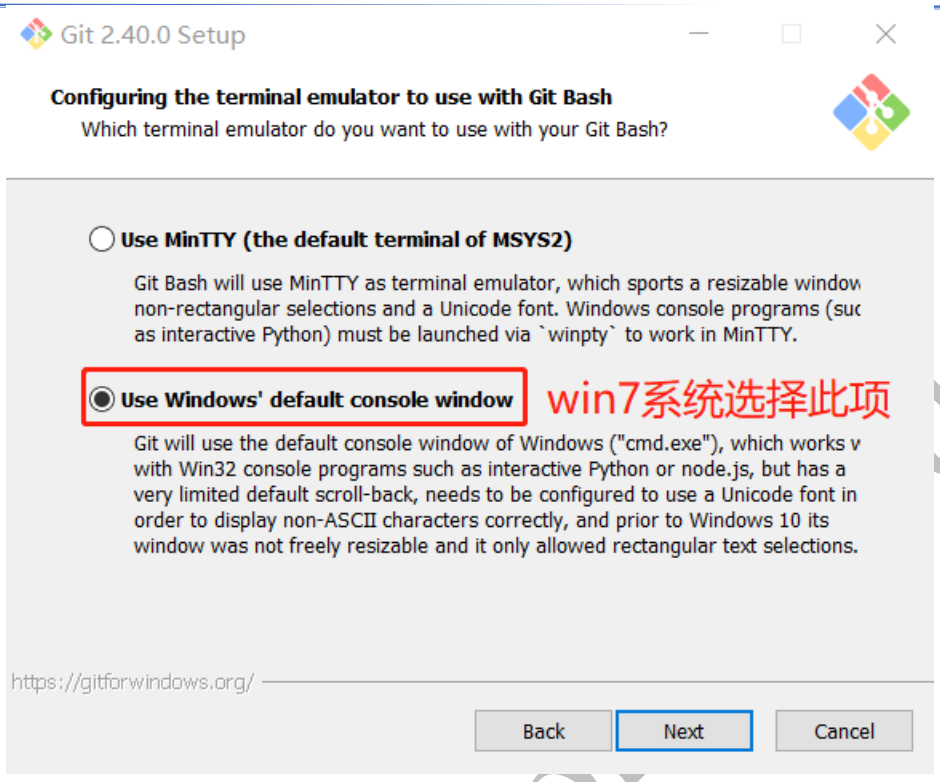
Git 安装时需特别注意下面几点！

windows10 环境,，需勾选如下选项：

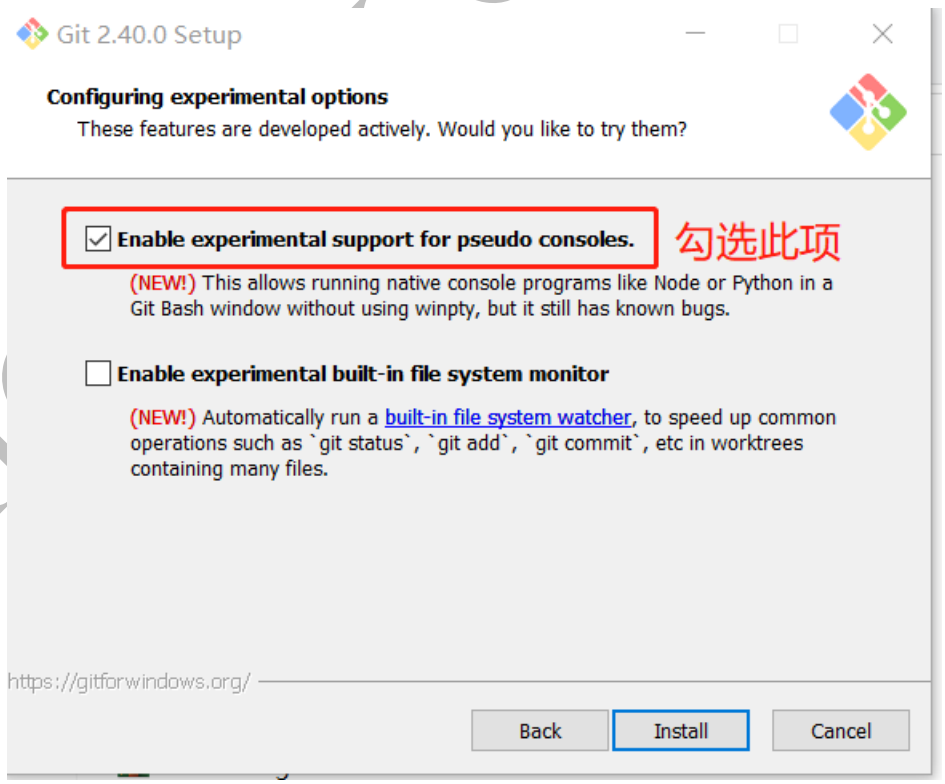


windows7 环境， 需勾选如下选项：

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windows7 和 window10 环境， 都需勾选如下选项：



安装完成后，检查 Git 版本信息：

```
comuser@ip-AC10022A MINGW64 /p/00_development/comuser/code
$ git --version
git version 2.28.0.windows.1
```

1.3.Python

Python 用做 SDK 脚本处理，软件版本需 3.8.10 及以上。可从如下路径获取安装包：

“SDK/tools/python/”*

1.3.1.Environment variable settings

安装时勾选添加系统环境变量。



安装完成后，检查软件版本信息：

```
comuser@ip-AC10022A MINGW64 /p/00_development/comuser/code
$ python --version
Python 3.8.10
```

1.3.2.Python libs

SDK 需要安装依赖部分必须库，可通过如下命令安装：

pip install colorama

pip install kconfiglib

pip install pandas

pip install prettytable

pip install numpy

pip install openpyxl

pip install cryptography

pip install click

pip install windows-curses

所有必须的库文件也在 *"SDK/tools/python/libs"* 目录中，也可直接安装本地库:

注意: **numpy-1.24.2-cp38-cp38-win_amd64.whl** 必须在 **pandas-1.5.3-cp38-cp38-win_amd64.whl** 之前安装！

pip install colorama-0.4.6-py2.py3-none-any.whl

pip install kconfiglib-14.1.0-py2.py3-none-any.whl

pip install numpy-1.24.2-cp38-cp38-win_amd64.whl

pip install pandas-1.5.3-cp38-cp38-win_amd64.whl

pip install prettytable-3.6.0-py3-none-any.whl

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pip install openpyxl-3.1.2-py2.py3-none-any.whl

pip install cryptography-39.0.2-cp36-abi3-win_amd64.whl

pip install click-8.1.3-py3-none-any.whl

pip install windows_curses-2.3.1-cp38-cp38-win_amd64.whl

安装时若提示 pip 报错，需要更新 pip，更新方法如下：

```
$ pip install colorama
Requirement already satisfied: colorama in c:\users\administrator\appdata\local\programs\python\python38\lib\site-packages (0.4.6)
WARNING: You are using pip version 21.1.1; however, version 23.0.1 is available.
You should consider upgrading via the 'c:\users\administrator\appdata\local\programs\python\python38\python.exe -m pip install --upg
rade pip' command.
```

Python update pip command(by network):

python -m pip install –upgrade pip

Python update pip command(by local .whl):

python -m pip install pip-23.0.1-py3-none-any.whl

1.4.Ninja

Ninja 为加速编译工具，工具路径如下：

“SDK/tools/ninja/”*

需添加 ninja 到环境变量的系统“path”中:

XXX\ninja

完成后，检查 ninja 版本信息：

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```
comuser@ip-AC10022A MINGW64 /p/00_development/comuser/code
$ ninja --version
1.10.2
```

1.5. Toolchain

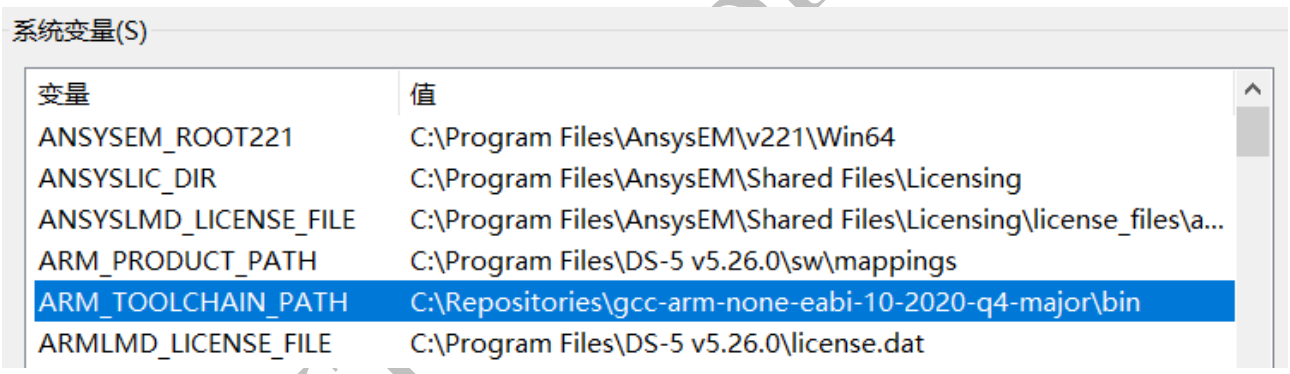
SDK 中包括编译工具链压缩包，路径如下，请将其解压至本地路径：

“SDK/tools/gcc-arm-none-eabi-10-2020-q4-major.rar”

添加编译工具链环境变量：

ARM_TOOLCHAIN_PATH=xxx\gcc-arm-none-eabi-10-2020-q4-major\bin

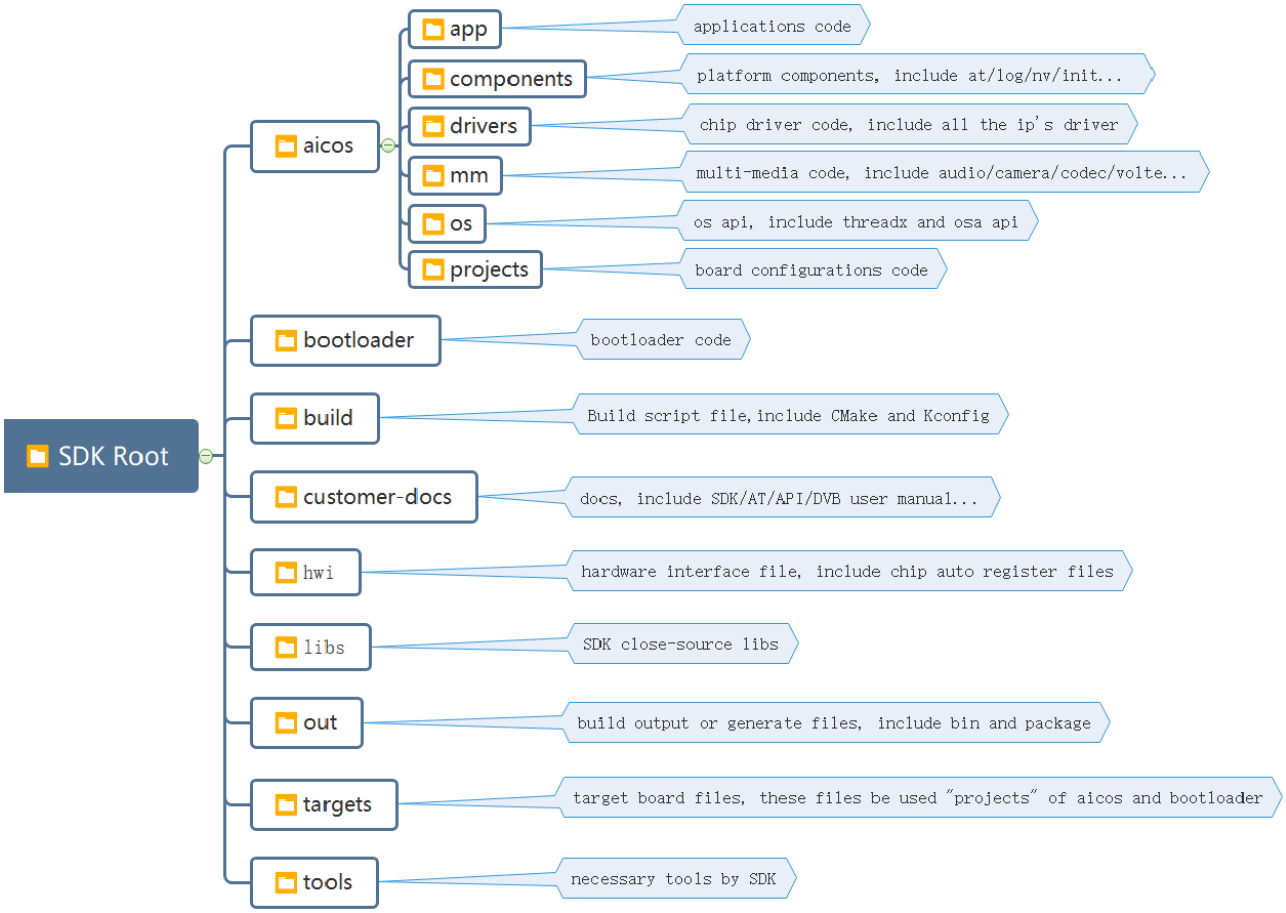
For example:



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2. 目录结构

SDK目录结构如下:



1. all the necessary tools could be found in "tools" folder which needed by chapters 1
2. all the necessary script files or commands could be found in "build" folder which needed by chapters 3
3. the user manual docs in "customer-docs" folder
4. "libs" folder includes all the sdk close-source libs
5. "out" folder includes all the output or generated files after build complete
6. "aicos/app" folder includes all the applications code
7. "aicos/components" folder includes platform components, e.g. at/log/nv/init/diag/gps/console …
8. "aicos/drivers" folder includes all the chip ip's driver code
9. "aicos/mm" folder includes all the multi media code, e.g. audio/camera/codec/volte …
10. "aicos/os" folder includes all the system api code, include threadx and osa api
11. "aicos/projects" folder includes all the board configurations code

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3. 编译方法

SDK目录结构如下，上章节有详细说明：

aicos	2022/11/5 11:08	文件夹
bootloader	2022/11/5 11:08	文件夹
build	2022/11/5 11:08	文件夹
customer-docs	2022/11/5 10:50	文件夹
hwi	2022/11/5 11:08	文件夹
libs	2022/11/5 11:07	文件夹
out	2022/11/5 11:14	文件夹
targets	2022/11/3 10:19	文件夹
tools	2022/11/5 11:08	文件夹

SDK编译都是在根目录下执行，编译的脚本都放在build目录下。

3.1. Prebuild

3.1.1. Terminal

SDK编译支持两种终端: Git Bash 和Windows cmd.

- Git Bash

建议且推荐使用Git Bash编译，命令如下：

python build/env.py or ./build/env.py

```
$ ./build/env.py
=========================  aic support boards  =========================
--  0 gx316c_h3                             /* 2+4 Tracker board */
--  1 gx316h_h1                             /* 8+8 board */
--  2 gx316h_h1_sec                         /* 8+8 security board */
--  3 gx316h_h2                             /* 8+8 watch board */
--  4 gx316h_h3                             /* 8+8 feature phone board */
--  5 gx316l_h1                             /* 4+4 board */
--  6 gx316l_h2                             /* 4+4 cloud horn board */
```

- Windows cmd

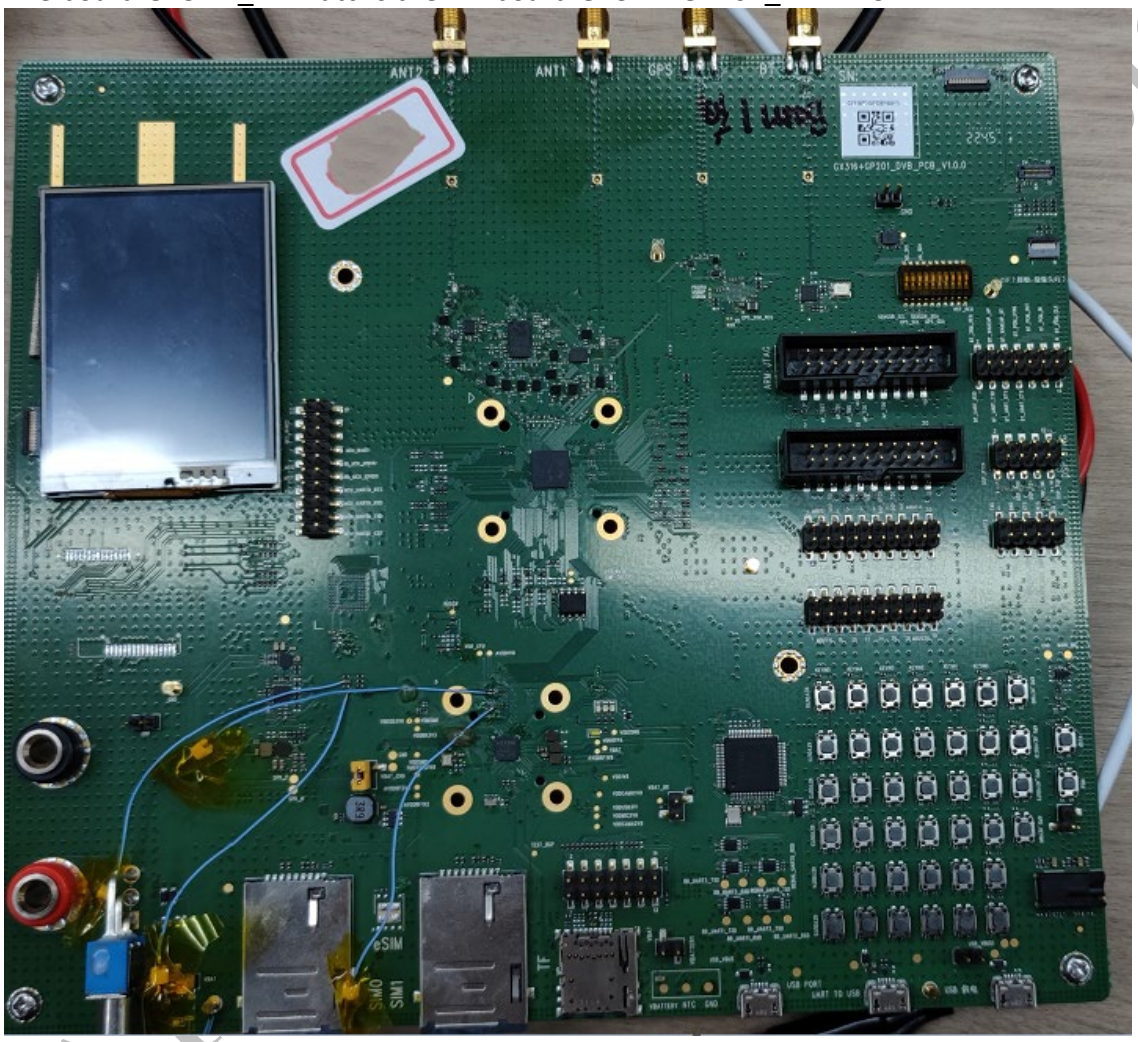
Windows cmd同样支持，命令如下:

python build\env.py or build\env.py

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```
C:\Users\yongfu.long\Downloads\sdk>build\env.py
==========================  aic support boards  ==========================
--  0 gx316c_h3                                 /* 2+4 Tracker board */
--  1 gx316h_h1                                 /* 8+8 board */
--  2 gx316h_h1_sec                             /* 8+8 security board */
--  3 gx316h_h2                                 /* 8+8 watch board */
--  4 gx316h_h3                                 /* 8+8 feature phone board */
--  5 gx3161_h1                                 /* 4+4 board */
```

The board GX31Xl_h1 matchs the hw board GX31X+GP201_DVB+PCB:



3.2. Lunch

Lunch目录是用于选择不要编译的board。

选择章节3.1中list出来board，命令如下：

./lunch.py <board_name or board_num>

如，选择第一个board， “./lunch.py 0” as below:

```
comuser@ip-AC10022A MINGW64 /p/00_development/comuser/code
$ ./lunch.py 0
====== aic lunch board ======
-- lunched: gx3161_h1
--   |
-- board info:
--             chip        -> genesis
--             ap board    -> gx3161_h1
--             cp board    -> gx3161_h1
--             adsp board  -> gx3161_h1
--             bl board    -> gx316
--             tfm board   -> NA
```

3.3. Make help

命令支持help帮助信息，命令如下：

./make.py -h or **./make.py --help**

```
$ ./make.py --help
====================build command help====================
Usage:./make.py [cmd] [oprtion]                           --
Enumerate all the used commands:                          --
-- ap               build aicos ap subsys                 --
-- cp               build aicos cp subsys                 --
-- adsp             build aicos adsp subsys               --
-- bl               build bootloader                      --
-- ap menuconfig    build aicos ap subsys kconfig         --
-- cp menuconfig    build aicos cp subsys kconfig         --
-- ap kuconfig      build aicos ap subsys kconfig         --
--                  and copy generated config para to     --
--                  board default_config file             --
-- cp kuconfig      build aicos cp subsys kconfig         --
--                  and copy generated config para to     --
--                  board default_config file             --
-- ap guiconfig     build aicos ap subsys guiconfig       --
-- cp guiconfig     build aicos cp subsys guiconfig       --
-- all              build all the software target, include--
--                  ap, cp, dsp, tfm, bootloader e.g.     --
-- clean            clean the target board build output   --
--                  files.                                --
-- idx              do as same as idx update.             --
-- idx update       update the new trace to id and update --
--                  and update the string lib files.      --
-- idx rebuild      Rebuild all the trace to id and       --
--                  and rebuild all the string lib files. --
```

3.4. Make subsys

编译支持单独编译某个subsys，命令如下:

./make.py <subsys_name>

如，单独编译 “ap” ， “./make.py ap”:

```
comuser@ip-AC10022A MINGW64 /p/00_development/comuser/code
$ ./make.py ap
================================
-- build gx316l_h1 ap start
================================
nvram info: aic nvram compiler all json file end, time cost:0.12s
Parsing P:/00_development/comuser/code/aicos/Kconfig
Loaded configuration 'P:/00_development/comuser/code/out/gx316l_h1/ap/.config'
No change to configuration in 'P:/00_development/comuser/code/out/gx316l_h1/ap/.config'
No change to Kconfig header in 'P:/00_development/comuser/code/out/gx316l_h1/ap/auto_generate/kconfig/include/generated/
autoconf.h'
-- Found toolchain: gnuarmemb (C:/Repositories/gcc-arm-none-eabi-10-2020-q4-major/bin/..)
-- Using default tx_user.h file
```

libname	text	bss	data	rodata	RO(text+rodata)	RW(bss+data)	lib_total
['libmm.a']	11658	1384	288	64024	75682	1672	77354
['libm.a', 'lib_a']	5064	0	0	472	5536	0	5536
['libdrivers.a']	49510	47056	3547	16109	65619	50603	116222
['libcodec.a']	44	8	0	10	54	8	62
['libaic_osa.a']	10182	72974	2	1480	11662	72976	84638
['libthreadx.a']	22488	2512	4	112	22600	2516	25116
['libnetxduo.a']	78040	17772	166	1855	79895	17938	97833
['libusbx.a']	12078	1004	209	690	12768	1213	13981
['libboard.a']	156	0	236	69	225	236	461
['libaic_components.a']	62254	11834	943	25733	87987	12777	100764
['libaic_nvram.a']	4438	8	120	3085	7523	128	7651
['libaic_miscdata.a']	7040	4	0	2722	9762	4	9766
['libfs.a']	5354	16	436	4916	10270	452	10722
['libfilex.a']	50554	876	111	37	50591	987	51578
['liblevelx.a']	4726	8	0	108	4834	8	4842
['libc_nano.a', 'lib_a']	280	0	100	257	537	100	637
['libgcc.a']	3256	0	0	0	3256	0	3256
['libaic_mcx.a']	576	16009	1	380	956	16010	16966
Total Size:	327698	171465	6163	122059	449757	177628	627385

```
================================
build gx316l_h1 ap end, time cost:3.77s
================================
```

3.5. Make all

全工程编译，该命令编译所有项目，并生成对应bin，打包成可执行pkg存放于out目录对应位置。(e.g. root/out/GX31XI_h1/GX31XI_h1.pkg) :

./make.py all

```
comuser@ip-AC10022A MINGW64 /p/00_development/comuser/code
$ ./make.py all
-- make all the gx316l_h1 os binary
-- build gx316l_h1 ap start
nvram info: aic nvram compiler all json file end, time cost:0.12s
Parsing P:/00_development/comuser/code/aicos/Kconfig
Loaded configuration 'P:/00_development/comuser/code/out/gx316l_h1/ap/.config'
No change to configuration in 'P:/00_development/comuser/code/out/gx316l_h1/ap/.config'
No change to Kconfig header in 'P:/00_development/comuser/code/out/gx316l_h1/ap/auto_generate/kconfig/include/generated/
autoconf.h'
-- Found toolchain: gnuarmemb (C:/Repositories/gcc-arm-none-eabi-10-2020-q4-major/bin/..)
-- Using default tx_user.h file
```

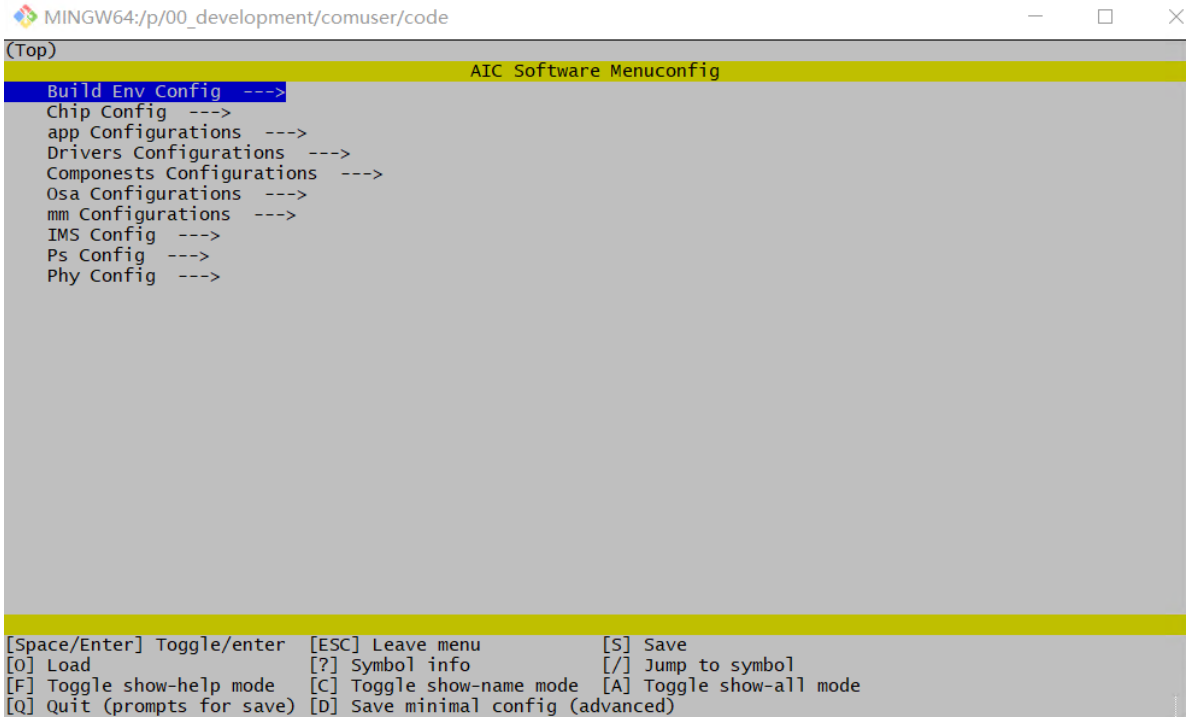
```
-- build gx316l_h1 bl end
-- make all the gx316l_h1 os binary done
 ==== Script Version: 1.0 ====
P:\00_development\comuser\code\aicos/projects/genesis/pkg/bin_cfg.json
P:\00_development\comuser\code\aicos/projects/genesis/pkg/bin_cfg.json
        Making P:\00_development\comuser\code\out\gx316l_h1/gx316l_h1.pkg ...
        packing P:\00_development\comuser\code\out\gx316l_h1\bootloader\bin\gx316_dl_signed.bin ...
        packing P:\00_development\comuser\code\out\gx316l_h1\bootloader\bin\gx316_bl_signed.bin ...
        packing P:\00_development\comuser\code\out\gx316l_h1\ap\bin\aicos_ap.bin ...
        packing P:\00_development\comuser\code\out\gx316l_h1\cp\bin\aicos_cp_flash.bin ...
        packing P:\00_development\comuser\code\out\gx316l_h1\cp\bin\aicos_cp_ram_compressed.bin ...
        packing P:\00_development\comuser\code\out\gx316l_h1\adsp\bin\dtcm_compressed.bin ...
        packing P:\00_development\comuser\code\out\gx316l_h1\adsp\bin\ptcm_compressed.bin ...
        packing P:\00_development\comuser\code\out\gx316l_h1\ap\bin\aic_ap_miscdata.bin ...
        packing P:\00_development\comuser\code\out\gx316l_h1\cp\bin\aic_nvram_data.bin ...

----- SUCCESS -----
pkg: P:\00_development\comuser\code\out\gx316l_h1/gx316l_h1.pkg
-- gx316l_h1 makepkg done
```

3.6. Make menuconfig

./make.py <subsys> menuconfig

使用menuconfig的图形界面对宏进行可视化配置，通过键盘即可完成对项目的所有Kconfig宏配置，该命令生成.config会在out目录中，projects对应board目录下的default文件不会被修改。



3.7. Make kuconfig

./make.py <subsys> kuconfig

Kuconfig功能和menuconfig类似，menuconfig功能如章节3.6所述，而kuconfig会将其修改的宏配置修改保存至projects对应board目录下的default文件中：

```
-- Build files have been written to: P:/00_development/comuser/code/out/gx316l_h1/ap
[0/1] cmd.exe /C "cd /D P:\00_development\comuser\code\...config.py P:/00_development/comuser/code/aicos/Kconfig"
Loaded configuration 'P:/00_development/comuser/code/out/gx316l_h1/ap/.config'
No changes to save (for 'P:/00_development/comuser/code/out/gx316l_h1/ap/.config')
New config copy to file P:\00_development\comuser\code\aicos\projects\genesis\ap\gx316l_h1\default_config
kconfig done!
```

3.8. Make clean

./make.py clean

命令用于删除编译生成的文件。

```
comuser@ip-AC10022A MINGW64 /p/00_development/comuser/code
$ ./make.py clean
-- clean now
-- clean complete
```

3.9.编译常见问题

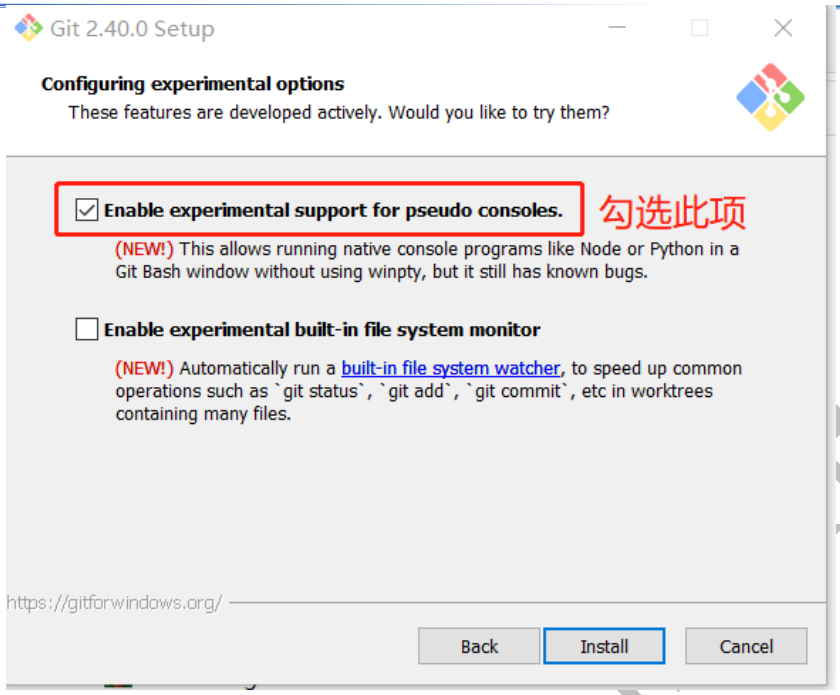
3.9.1. make ap menuconfig时报错

```
Configuration saved to 'D:/code/03_sdk/sibeitu/aicos/out/gx316h_h1/ap/.config'
Kconfig header saved to 'D:/code/03_sdk/sibeitu/aicos/out/gx316h_h1/ap/auto_generate/kconfig/include/generated/autoconf.h'
-- Found toolchain: gnuarmemb (C:/toolchain/gcc-arm-none-eabi-10-2020-q4-major/bin/..)
-- The C compiler identification is GNU 10.2.1
-- The CXX compiler identification is GNU 10.2.1
-- Detecting C compiler ABI info
-- Detecting C compiler ABI info - done
-- Check for working C compiler: C:/toolchain/gcc-arm-none-eabi-10-2020-q4-major/bin/arm-none-eabi-gcc.exe - skipped
-- Detecting C compile features
-- Detecting C compile features - done
-- Detecting CXX compiler ABI info
-- Detecting CXX compiler ABI info - done
-- Check for working CXX compiler: C:/toolchain/gcc-arm-none-eabi-10-2020-q4-major/bin/arm-none-eabi-g++.exe - skipped
-- Detecting CXX compile features
-- Detecting CXX compile features - done
-- The ASM compiler identification is GNU
-- Found assembler: C:/toolchain/gcc-arm-none-eabi-10-2020-q4-major/bin/arm-none-eabi-gcc.exe
-- Configuring done
-- Generating done
-- Build files have been written to: D:/code/03_sdk/sibeitu/aicos/out/gx316h_h1/ap
[0/1] cmd.exe /C "cd /D D:\code\03_sdk\sibeitu\aicos\out\gx316h_h1\ap\auto_generate\kconfig && "C:\Program Files\CMake\bin\cmake.exe" -E env
 PYTHON_EXECUTABLE=C:/tools/python311/python.exe srctree=D:/code/03_sdk/sibeitu/aicos/aicos KCONFIG_BASE=D:/code/03_sdk/sibeitu/aicos/aicos
KCONFIG_CONFIG=D:/code/03_sdk/sibeitu/aicos/out/gx316h_h1/ap/.config ARCH= CMAKE_BINARY_DIR=D:/code/03_sdk/sibeitu/aicos/out/gx316h_h1/ap AR
CH_DIR= PROJECT_NAME=gx316h_h1 C:/tools/python311/python.exe -B D:/code/03_sdk/sibeitu/aicos/build/kconfig/menuconfig.py D:/code/03_sdk/sibe
itu/aicos/aicos/Kconfig"

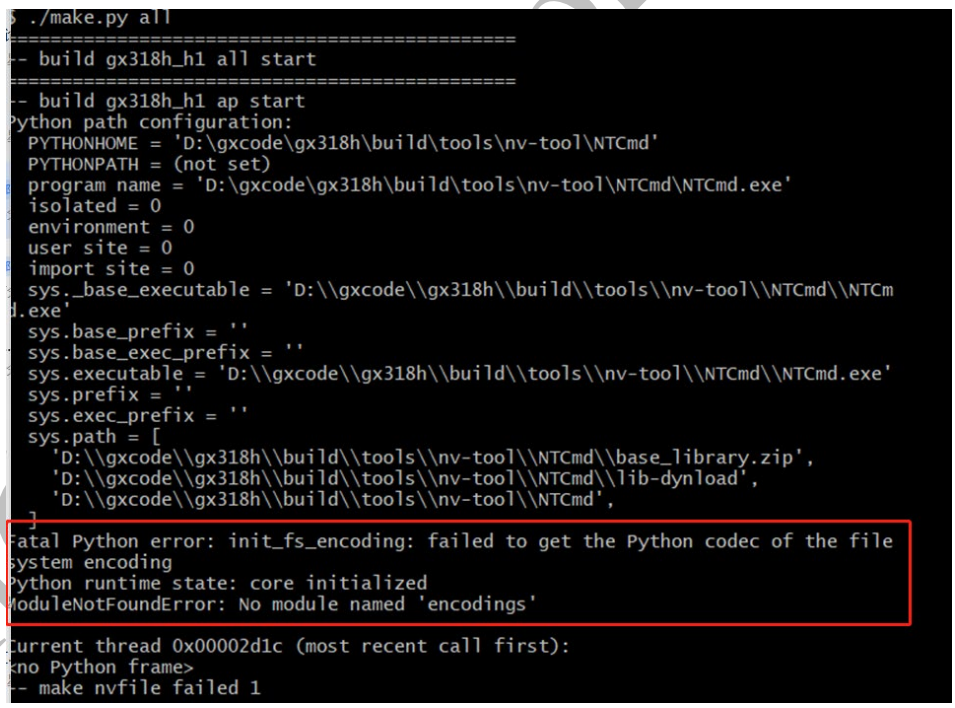
Redirection is not supported.
FAILED: CMakeFiles/menuconfig
cmd.exe /C "cd /D D:\code\03_sdk\sibeitu\aicos\out\gx316h_h1\ap\auto_generate\kconfig && "C:\Program Files\CMake\bin\cmake.exe" -E env PYTHO
N_EXECUTABLE=C:/tools/python311/python.exe srctree=D:/code/03_sdk/sibeitu/aicos/aicos KCONFIG_BASE=D:/code/03_sdk/sibeitu/aicos/aicos KCONFI
G_CONFIG=D:/code/03_sdk/sibeitu/aicos/out/gx316h_h1/ap/.config ARCH= CMAKE_BINARY_DIR=D:/code/03_sdk/sibeitu/aicos/out/gx316h_h1/ap ARCH_DIR
= PROJECT_NAME=gx316h_h1 C:/tools/python311/python.exe -B D:/code/03_sdk/sibeitu/aicos/build/kconfig/menuconfig.py D:/code/03_sdk/sibeitu/ai
cos/aicos/Kconfig"
ninja: build stopped: subcommand failed.
-- make menuconfig failed
```

此报错是因为安装git bash时有选项未勾选，可在章节1.2中找到解决方法：

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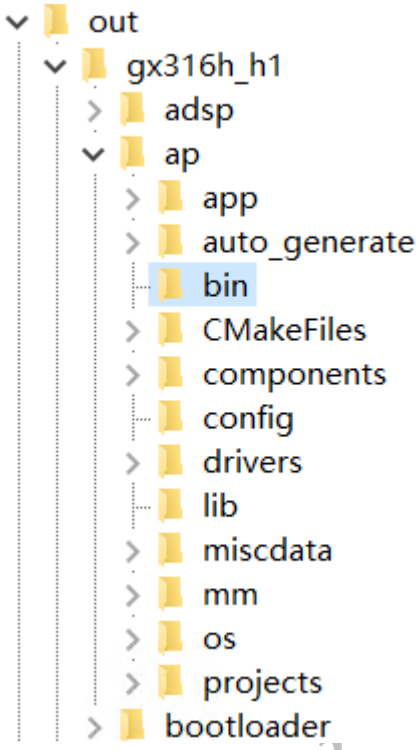
3.9.2.调用python时报encoding错



导致该错误的原因是本地环境的杀毒软件或防火墙阻止的部分软件执行。
解决方法：关闭杀毒软件或者将归芯的nv等脚本软件加入到白名单。

4. 生成文件

编译文件会生成在out对应的board目录下：

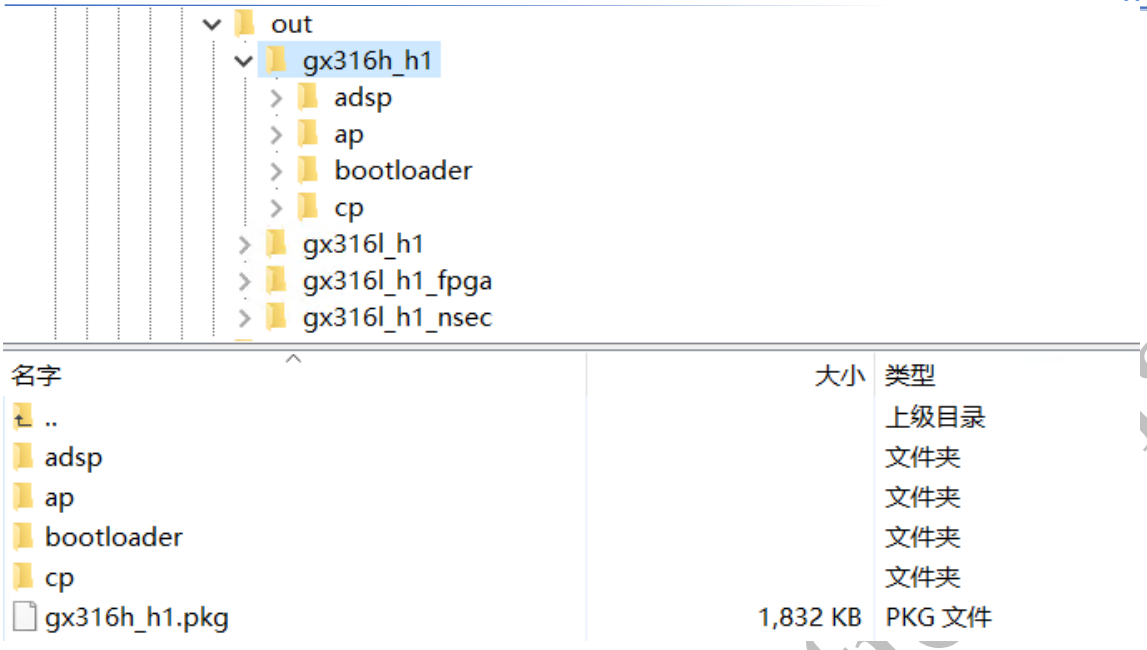


生成的image、bin、elf等文件在其对应的“bin”目录，如:

aic_ap_miscdata.bin	1 KB	BIN 文件	2022/8/18 19:49:43
aicos_ap.bin	460 KB	BIN 文件	2022/8/18 19:50:08
aicos_ap.elf	7,807 KB	ELF 文件	2022/8/18 19:50:08
aicos_ap.map	2,345 KB	Linker Address Map	2022/8/18 19:50:08

生成的pkg在board的根目录下：

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5. 代码维护

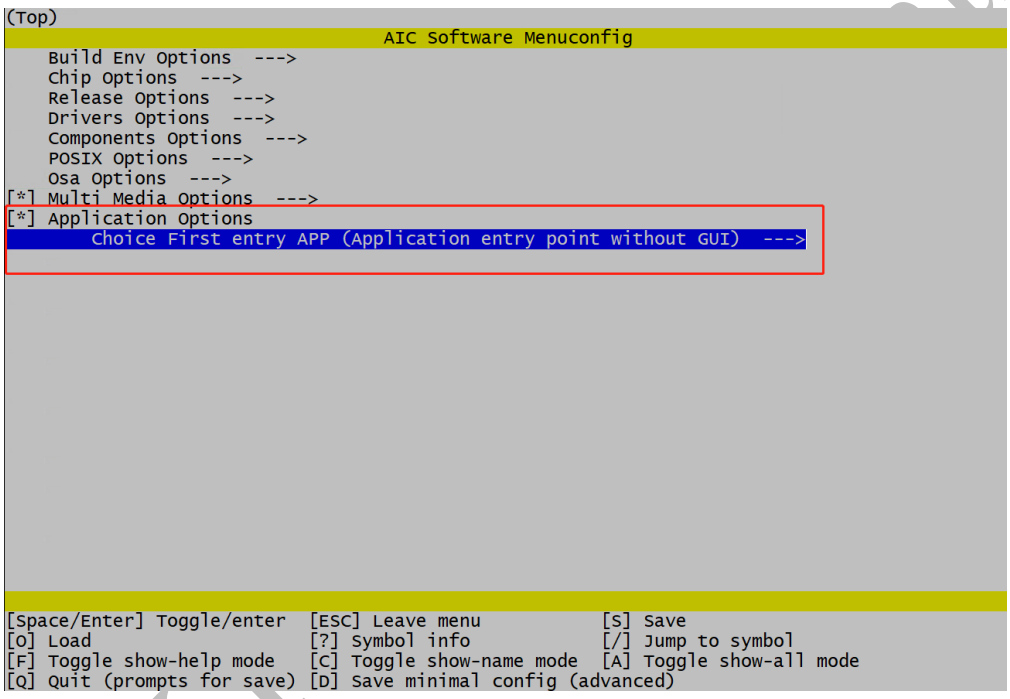
本章节主要介绍SDK code相关代码维护功能.

5.1. application代码入口

SDK代码中给用户提供了一个application入口，该功能可在menuconfig中动态使能，命令如下：

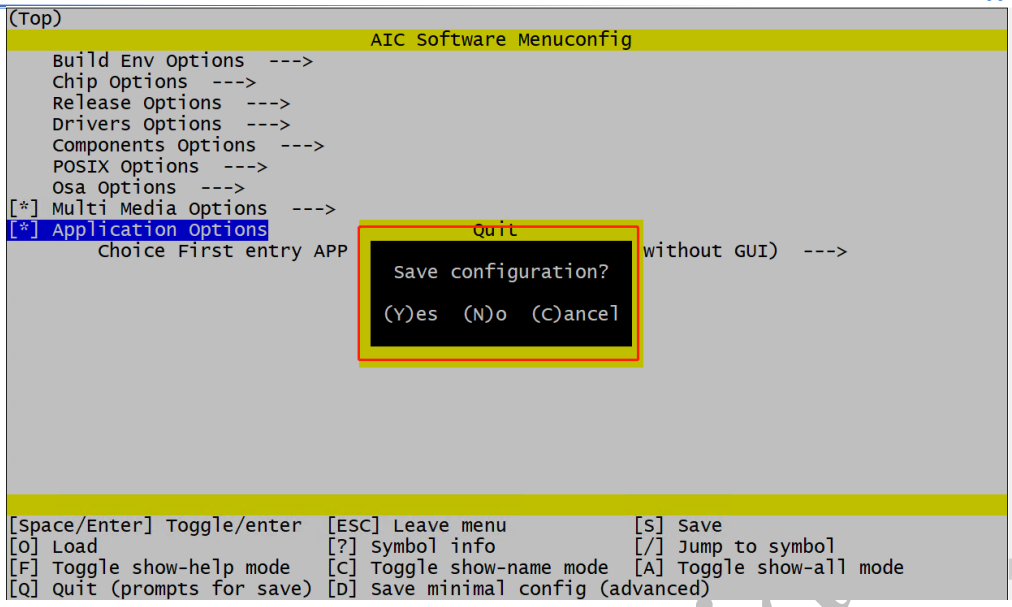
```
./make.py ap kuconfig
```

选择“Application entry point without GUI” :



保存修改配置:

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然后执行 “./mak.py all” 生成新的pkg:

./make.py all

application entry入口函数为**AppEntryWithoutGUI**；路径：SDK/aicos/app/app.c.

```
/* Entry point for application, custom this function to meet your requirement */
static void AppEntryWithoutGUI(ULONG thread_input)
{
    /* Initial API of APP */
    //app_demo_init();

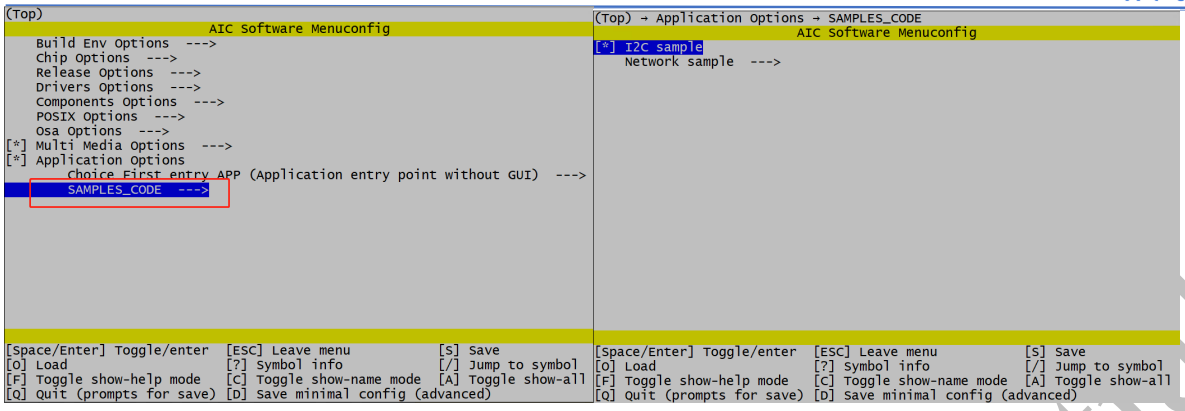
    while (1) {
#ifndef AIDX_AUTO_GEN
        alog_info("Application entry point!\n");
#else
        /* Auto gen by tool, don't change it. */
        aidx_info(AIC_THIS_FILE, 0x1adb43, "");
#endif
        /* 15 seconds */
        aos_sleep(15000);
    }
}
```

5.2. Sample code

示例code路径如下:

“SDK/app/samples/”*

所有sample code可以通过命令“./make.py ap menuconfig” 选择和使能, 然后通过 “./make.py all” 生成新的pkg已调试使用.



注意！sample code仅做调试参考使用，正式代码请关闭。

5.3. 添加source code

添加source code编译，需修改 CMakeLists.txt 文件.
如，添加 “custom_code.c” 到 “custom” 目录，修改如下:

```
# General configuration options
#
# Copyright 2022 by AIC Micro Corporation. All rights reserved.

aic_library_sources_ifdef(CONFIG_DRIVER_CUSTOM
    ${CMAKE_CURRENT_LIST_DIR}/custom_code.c
)
```

“CONFIG_DRIVER_CUSTOM” 是宏定义，仅在定义该宏时才将该c code加入到编译；
“CONFIG_DRIVER_CUSTOM” 是在KConfig中定义，在make kuconfig中被选中打开。

添加头文件路径，修改如下:

```
aic_include_directories(
    ./inc
)
```

如何确认文件是否被编译到，步骤如下:

1. 确保编译全部成功，没有报错
2. 在 out 目录下找到“build.ninja”
3. 搜索 “custom_code.c.obj”字串，如果找到说明被编译，否则没有

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5.4. 添加模块

添加一个模块到sdk中，需要修改Kconfig 和CMakeLists.txt 文件.

如，添加 “custom” 到“drivers”目录。

- Kconfig

新建一个Kconfig文件， 按照如下方式添加宏定义:

```
# General configuration options
#
# Copyright 2022 by AIC Micro Corporation. All rights reserved.

config DRIVER_CUSTOM
    bool "Custom driver"
    default n
    help
      Define the Custom IP, descripte the IP functions

config DRIVER_CUSTOM_VER
    string "Custom driver IP version"
    depends on DRIVER_CUSTOM
    default "cstm"
    help
      Custom IP version, different projects would use different version
      of custom IP.
```

然后在父级目录中引用，再用 “./make.py <subsys> kuconfig” 选中该模块

```
(Top) → Drivers Configurations
                                    AIC Software Menuconfig
[ ] ADRMAP module
[ ] ADSP boot code
[*] aic mailbox driver
    Select mailbox driver (aic mailbox ip version r1px)  --->
[*] enable ambox phy
[*] BusMonitor module
[*] Performance Monitor
(r1p0) Bus monitor driver IP version
[ ] AIC Clock
    Select driver type  ----
[*] Coresight driver
[*]     MTB driver
    Crypto Component Configurations  --->
[*] Custom driver
(cstm)  Custom driver IP version (NEW)
[*] Driver DMA
(r1p0)  DMA IP version
    Efuse Configurations  --->
[*] Driver EIPU module
[ ] Driver example
    Select example driver version  ----
[*] Flash Managemant Device (FMD) support  --->
[*] GPD driver
    ↓↓↓↓↓↓↓↓↓↓↓↓↓
[Space/Enter] Toggle/enter  [ESC] Leave menu          [S] Save
[O] Load                    [?] Symbol info           [/] Jump to symbol
[F] Toggle show-help mode   [C] Toggle show-name mode  [A] Toggle show-all mode
[Q] Quit (prompts for save) [D] Save minimal config (advanced)
```

- CMakeList.txt

在父级目录添加 CMakeList 的引用，如下：

```
add_subdirectory(arch/${ARCH})
add_subdirectory(sleep)
add_subdirectory(sound)
add_subdirectory(usb)
add_subdirectory(iodev)
add_subdirectory(misc)
add_subdirectory(graphics)

add_subdirectory_ifdef(CONFIG_DRIVER_AMBOX          ambox)
add_subdirectory_ifdef(CONFIG_DRIVER_BUSMON         busmonitor)
add_subdirectory_ifdef(CONFIG_AIC_CLK               clk)
add_subdirectory_ifdef(CONFIG_DRIVER_CORESIGHT      coresight)
add_subdirectory_ifdef(CONFIG_DRIVER_CUSTOM         custom)
add_subdirectory_ifdef(CONFIG_DRIVER_DMA            dma)
add_subdirectory_ifdef(CONFIG_DRIVER_EXAMPLE        example)
add_subdirectory_ifdef(CONFIG_FMD                   fmd)
add_subdirectory_ifdef(CONFIG_DRIVER_GPD            gpd)
add_subdirectory_ifdef(CONFIG_DRIVER_GPIO           gpio)
add_subdirectory_ifdef(CONFIG_DRIVER_LZMA_DEC       lzma)
add_subdirectory_ifdef(CONFIG_NET_DRIVER            network)
```

在被添加的目录中新建一个CMakeList.txt，然后按需修改，如：

```
aic_library_sources_ifdef(CONFIG_DRIVER_CUSTOM
    ${CMAKE_CURRENT_LIST_DIR}/custom_code.c
)

aic_include_directories(
    ./inc
)
```

5.5. 添加3rd-party library

使用接口aic_library_import添加 3rd-party library.
e.g.
aic_library_import(3rd-party ${root}/path/lib3rd-party.a)

5.6. project目录下头文件overlap功能

在aicos/projects/genesis/ap目录里，支持头文件overlap功能，[board]/config目录下的头文件会覆盖common_inc目录下的头文件，目的是使board下头文件相对更好维护，只需要在有差异的board下添加一个差异的头文件，其他board则使用common_inc目录下的头文件。

如：在common_inc下有customer_header.h头文件，所有board都会使用，但gx316l_h4会有差异，则仅需在gx316l_h4/config目录下再添加一个customer_header.h，并修改成gx316l_h4即可，无需每个board都修改。

5.7.第三方编译库文件

编译工具链：

https://developer.arm.com/-/media/Files/downloads/gnu-rm/10-2020q4/gcc-arm-none-eabi-10-2020-q4-major-win32.zip?revision=ffcaa06c-940d-4319-8d7e-d3070092d392&la=en&hash=130196DDF95B0D9817E1FDE08A725C7EBFBFB5B8

编译选项：

FLAGS = -Os -ffreestanding -fno-common -g -gdwarf-4 -fdiagnostics-color=always -mcpu=cortex-m33 -mthumb -mabi=aapcs --specs=nano.specs -Wall -Wformat -Wformat-security -Wno-format-zero-length -Wno-main -Werror -Wno-pointer-sign -Wpointer-arith -Wexpansion-to-defined -Wno-address-of-packed-member -Wno-unused-but-set-variable -Werror=implicit-int -ffunction-sections -fdata-sections -fno-pie -fno-pic -fno-builtin -Wno-format -Wno-format-security

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6. Release 版本编译

当软件版本趋于稳定，达到量产状态，基本功能都正常，且没有严重assert和死机问题时，则需要编译release版本，该版本的编译方法如下（其他命令无差异）：

```
./make.py all user
```

编译出来的release版本和debug版本差异如下：

- Watchdog超时后将直接重启
- Log功能默认关闭，AP/CP/DSP 默认不会输出
- Assert之后不再做dump，将直接重启
- JTAG禁用
- Bus monitor及其他芯片debug功能将关闭

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7. Board维护

归芯软件中存在多个board，目的是为了适配不同硬件差异。Board差异的部分主要体现在下面几个部分：

- target子目录下的jason文件
- target\genesis目录下的board文件夹
- bootloader\projects下的board文件夹
- bootloader\build\libs下的board文件夹
- aicos\libs下的board文件夹
- aicos\projects\genesis\ap目录下的board文件夹
- aicos\projects\genesis\nvram\ap目录下的nv board文件夹
- aicos\projects\genesis\nvram\cp目录下的nv board文件夹

7.1. Jason文件及对应文件夹说明

```
"Board info": {
    "genesis": [
        {
            "project board name":    "gx316h_h1",
            "ap board name":         "gx316h_h1",
            "sdk base board name":   "gx316h_h1",
            "md board name":         "gx316h_h1",
            "cp board name":         "gx316h_h1",
            "nv board name":         "gx316h_h1",
            "dsp board name":        "gx316h",
            "bootloader board name":"gx316h_h1",
            "tfm board name":        "NA",
            "SW version":            "0",
            "Comments":              "8+8 board"
        },
```

- project board name：需要创建的board 名字，其名字为独一无二，不可以复用。
  其体现在target\genesis目录下需要创建对应的文件夹，可以直接拷贝参考board文件夹，无需做任何修改
- ap board name：AP board名字，保存AP差异相关配置，一般客户都是AP 差异，需创建AP board
  其体现在aicos\projects\genesis\ap目录下的board文件夹，可以拷贝参考board文件夹，修改board差异部分的config
- sdk base board name：sdk的参考board，可以不定义该参数，不定义的情况下其值和project board name是一样
  其体现在aicos\libs目录下的board文件夹，若继承libs下现有的board，直接填入board名，无需在libs目录下创建文件夹；若新建board不定义该变量，则直接默认用project board name，就必须在libs目录下创建对应的文件夹
- md board name：ap差异相关的nv board，用来保存AP差异的nv数据，该参数可选，

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可不定义
其体现在aicos\projects\genesis\nvram\ap目录下的nv board文件夹，可以不定义该参数，不定义情况下使用aicos\projects\genesis\nvram\ap\common下的nv数据

- cp board name：CP board名字，保存CP 差异配置，客户不需要关心，直接拷贝即可
- nv board name：CP nv board，保存CP nv相关差异，该参数可选，可不定义
  其体现在aicos\projects\genesis\nvram\cp目录下的nv board文件夹，可以不定义该参数，不定义情况下使用aicos\projects\genesis\nvram\cp\common下的nv数据
- dsp board name：dsp board名字，客户不关心，直接拷贝即可
- bootloader board name：uboot board名字，保存uboot差异配置
  其体现在bootloader\projects目录下的board文件夹，如果uboot没有差异，直接拷贝或者直接填写参考board名字，则无需在bootloader下创建文件夹，若有差异，创建对应文件夹，修改差异部分配置，bootloader\build\libs下的board文件需要同步修改。
- Comments：其描述board信息，在./build/env.py和lunch board时显示
- 其他board信息（tfm board name、SW version）直接拷贝即可

7.2. 建board举例

举例一：创建仅有AP 差异的board（基础board gx316l_h4）

gx316l_h4参考board配置如下：

```
{
    "project board name":    "gx316l_h4",
    "ap board name":         "gx316l_h4",
    "md board name":         "gx316l_h4",
    "cp board name":         "gx316l_h4",
    "nv board name":         "gx316l_h4",
    "dsp board name":        "gx316l",
    "bootloader board name":"gx316l_h4",
    "tfm board name":        "NA",
    "SW version":            "0",
    "Comments":              "4+4 adult watch board"
},
```

创建一个board名字为gx316l_h4_new的文件夹，拷贝gx316l_h4所有文件，创建后修改文件夹下的board_info.jason，如下：

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```
{
    "project board name":    "gx316l_h4_new",
    "ap board name":         "gx316l_h4_new",
    "sdk base board name":   "gx316l_h4",
    "md board name":         "gx316l_h4",
    "cp board name":         "gx316l_h4",
    "nv board name":         "gx316l_h4",
    "dsp board name":        "gx316l",
    "bootloader board name":"gx316l_h4",
    "tfm board name":        "NA",
    "SW version":            "0",
    "Comments":              "gx316 new board"
},
```

所以主要工作如下：：

- ✧ targets\genesis下拷贝gx316l_h4，命名为gx316l_h4_new
- ✧ 修改如上代码到board的jason文件中
- ✧ aicos\projects\genesis\ap下拷贝gx316l_h4，命名为gx316l_h4_new，修改board相关差异即可

举例二：创建差异较多的board（基础board gx316l_h3）

例如差异包含：bootloader、ap、ap nv和cp nv

修改如下：

```
{
    "project board name":    "gx316l_h3_test",
    "ap board name":         "gx316l_h3_test",
    "sdk base board name":   "gx316l_h3",
    "md board name":         "gx316l_h3_test",
    "cp board name":         "gx316l_h3",
    "nv board name":         "gx316l_h3_test",
    "dsp board name":        "gx316l",
    "bootloader board name":"gx316l_h3_test",
    "tfm board name":        "NA",
    "SW version":            "0",
    "Comments":              "test board"
},
```

需要修改的部分如下：

- ➢ targets\genesis下拷贝gx316l_h3，命名为gx316l_h3_test
- ➢ 将如上修改添加到board的jason文件
- ➢ aicos\projects\genesis\ap下拷贝gx316l_h3，命名为gx316l_h3_test，修改board差异
- ➢ aicos\projects\genesis\nvram\ap下拷贝gx316l_h3，命名为gx316l_h3_test，修改board nv差异
- ➢ aicos\projects\genesis\nvram\cp下拷贝gx316l_h3，命名为gx316l_h3_test，修改board nv差异
- ➢ bootloader\projects下拷贝gx316l_h3，命名为gx316l_h3_test，修改board差异
- ➢ bootloader\build\libs下拷贝gx316l_h3，命名为gx316l_h3_test

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7.3. 新board编译

新建board或者对targets目录下的board_info.json文件做过任何修改，编译时需要执行章节3.1和章节3.2的预编译操作后再做”./make.py all”：

```
yongfu.long@szplatsrv04 MINGW64 ~/Downloads/sdk
$ ./build/env.py
=========================  aic support boards  =========================
--  0 gx316c_h3                              /* 2+4 Tracker board */
--  1 gx316h_h1                              /* 8+8 board */
--  2 gx316h_h1_sec                          /* 8+8 security board */
--  3 gx316h_h2                              /* 8+8 watch board */
--  4 gx316h_h3                              /* 8+8 feature phone board */
--  5 gx316l_h1                              /* 4+4 board */
--  6 gx316l_h2                              /* 4+4 cloud horn board */
--  7 gx316l_h3                              /* 2+4 Tracker board */
--  8 gx316l_h4                              /* 4+4 adult watch board */
--  9 gx318c_h1                              /* 2+4 gx318 board */
-- 10 gx318c_h3                              /* 2+4 Tracker board */
-- 11 gx318c_h4                              /* 2+4 gx318 adult watch board */
-- 12 gx318c_h5                              /* 2+2 gx318 adult watch board */
-- 13 gx318f_os                              /* 4+4 oversea board */
-- 14 gx318h_h1                              /* 8+8 gx318 board */
-- 15 gx318l_h4                              /* 4+4 gx318 adult watch board */

yongfu.long@szplatsrv04 MINGW64 ~/Downloads/sdk
$ ./lunch.py 8
=========================  aic lunch boards  =========================
-- lunched: gx316l_h4
--  |
-- board info:
--            chip        -> genesis
--            ap board    -> gx316l_h4
--            md board    -> gx316l_h4
--            cp board    -> gx31xl_volte
--            nv board    -> gx316l_h4
--            dsp board   -> gx316l
--            bl board    -> gx316l_h4
--            tfm board   -> NA
--            sw version  -> 0
--            information :  4+4 adult watch board

yongfu.long@szplatsrv04 MINGW64 ~/Downloads/sdk
$ ./make.py all
============================================
-- build gx316l_h4 all start
============================================
```

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8. patch修改及编译

客户拿到SDK后，可能会有归芯发过来的patch需要merge验证的情况，patch可能分多种，方法会有一定差异，patch主要有：

- ✧ 开源code及头文件
- ✧ 闭源.a库
- ✧ NV文件
- ✧ CP bin文件

注意！patch merge后，用章节3.1、3.2和3.5的全编译方法都是可以确保patch编译到package中。

8.1. 开源code及头文件

patch marge后，直接全编译即可。

```
$ ./build/env.py
==========================  aic support boards  ==========================
--  0 gx316c_h3                                 /* 2+4 Tracker board */
--  1 gx316h_h1                                 /* 8+8 board */
--  2 gx316h_h1_sec                             /* 8+8 security board */
--  3 gx316h_h2                                 /* 8+8 watch board */
--  4 gx316h_h3                                 /* 8+8 feature phone board */
--  5 gx316l_h1                                 /* 4+4 board */
--  6 gx316l_h2                                 /* 4+4 cloud horn board */
comuser@ip-AC10022A MINGW64 /p/00_development/comuser/code
$ ./lunch.py 0
====== aic lunch board ======
-- lunched: gx316l_h1
--  |
-- board info:
--             chip        -> genesis
--             ap board    -> gx316l_h1
--             cp board    -> gx316l_h1
--             adsp board  -> gx316l_h1
--             bl board    -> gx316
--             tfm board   -> NA
comuser@ip-AC10022A MINGW64 /p/00_development/comuser/code
$ ./make.py all
-- make all the gx316l_h1 os binary
-- build gx316l_h1 ap start
nvram info: aic nvram compiler all json file end, time cost:0.12s
Parsing P:/00_development/comuser/code/aicos/Kconfig
Loaded configuration 'P:/00_development/comuser/code/out/gx316l_h1/ap/.config'
No change to configuration in 'P:/00_development/comuser/code/out/gx316l_h1/ap/.config'
No change to Kconfig header in 'P:/00_development/comuser/code/out/gx316l_h1/ap/auto_generate/kconfig/include/generated/
autoconf.h'
-- Found toolchain: gnuarmemb (C:/Repositories/gcc-arm-none-eabi-10-2020-q4-major/bin/..)
-- Using default tx_user.h file
```

8.2. 闭源.a库

闭源库放在 [root]/libs/[board_name]/ap 目录下：

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libs › gx316l_h4 › ap

名称	修改日期	类型	大小
libaic_tele.a	2024/1/19 9:23	A 文件	1,840 KB
libaic_wifi.a	2024/1/19 9:23	A 文件	76 KB
libcs_components.a	2024/1/19 9:23	A 文件	701 KB
libcs_drivers.a	2024/1/19 9:23	A 文件	873 KB
libfilex.a	2024/1/19 9:23	A 文件	3,716 KB
liblevelx.a	2024/1/19 9:23	A 文件	303 KB
libnetxduo.a	2024/1/19 9:23	A 文件	18,805 KB
libthreadx.a	2024/1/19 9:23	A 文件	2,618 KB
libusbx.a	2024/1/19 9:23	A 文件	2,729 KB

拿到.a库后，替换到对应board下ap目录里的库，然后执行全编译即可生效。

8.3. NV文件

NV文件包括AP NV和CP NV文件，分别在（可参考章节7.1中nv board说明部分）：

aicos › projects › genesis › nvram

名称	修改日期	类型
ap	2024/1/19 14:12	文件夹
cp	2024/1/19 14:12	文件夹

将获取到的nv文件替换到对应目录，然后执行全编译即可生效。

8.4. CP bin文件

CP bin文件存放在[root]/libs/[board_name]/cp/bin目录下：

libs › gx316c_h3 › cp › bin

名称	修改日期	类型	大小
aic_nvram_data_native.bin	2024/1/19 9:12	BIN 文件	34 KB
aicos_cp_ram1.bin	2024/1/19 9:12	BIN 文件	401 KB
aicos_cp_ram1_compressed.bin	2024/1/19 9:12	BIN 文件	248 KB
aicos_cp_ram2.bin	2024/1/19 9:12	BIN 文件	335 KB
aicos_cp_ram2_compressed.bin	2024/1/19 9:12	BIN 文件	239 KB
aicos_cp_ram3.bin	2024/1/19 9:12	BIN 文件	374 KB
aicos_cp_ram3_compressed.bin	2024/1/19 9:12	BIN 文件	241 KB
aicos_cp_ram4.bin	2024/1/19 9:12	BIN 文件	539 KB
aicos_cp_ram4_compressed.bin	2024/1/19 9:12	BIN 文件	230 KB

将获取到的bin文件替换到对应目录，然后执行全编译即可生效。

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